
Harnessed Agentic Simulation for Controllable Decision Support under Uncertainty

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Abstract

This proposal studies a narrow problem: how an LLM-based agentic workflow can support financial portfolio scenario analysis without hiding assumptions, evidence, or intermediate reasoning. It does not aim to build a general autonomous agent or a financial prediction model. The proposed system, *Harnessed Agentic Simulation*, turns a portfolio decision question into a validated world specification, explicit evidence and assumptions, bounded simulation runs, provenance traces, and evidence-linked report claims. The central hypothesis is that such a workflow can improve traceability, verifiability, and auditability compared with one-shot LLM responses and simple prompt-only baselines. More broadly, the project asks how a long-horizon agentic workflow can remain stable when the model has only a limited context window.

1 Introduction

Financial decisions are often made with incomplete information. In portfolio analysis, a recommendation is only useful if the user can see which market assumptions were used, how risk was measured, and why one allocation is more robust than another. This makes traceability central rather than optional: investors and analysts need to inspect the evidence path behind a conclusion, check whether a result depends on a fragile assumption, and compare plausible futures before acting. Portfolio scenario analysis is therefore a suitable setting for decision support systems [8], decision analysis [6], and simulation-based reasoning.

Large language models can parse open-ended questions, summarize documents, and coordinate tools. However, a one-shot LLM response is weak for portfolio decisions because it usually merges evidence, assumptions, calculations, and recommendations into one fluent narrative. When the answer is wrong or unsupported, it is difficult to locate the failure point. This risk is related to hallucination and unsupported generation [7]. Existing agent work has advanced reasoning, acting, memory, coordination, and social simulation [11, 17, 18, 25, 29, 31], but it still leaves a practical gap for audited financial scenario analysis. This gap is especially important for long-running agents, where earlier assumptions and intermediate states may fall outside the active context window.

This proposal treats portfolio decision support as a staged workflow rather than a single answer-generation task. The system structures the decision, retrieves evidence, records assumptions, validates a WorldSpec, runs bounded scenarios, and generates claims that link back to evidence, assumptions, and trace events. The immediate research object is therefore precise: LLM-based agentic workflow design for traceable, verifiable, and auditable portfolio scenario analysis under uncertainty.

2 Research Gap and Questions

Existing LLM agent systems are usually optimized for task completion, dialogue, tool use, or benchmark performance. Classical agent-based modeling can represent interacting actors and emergent effects [3, 4, 15, 24], but traditional ABM often assumes that experts have already specified the model. Retrieval-augmented generation can ground answers in documents [10], and provenance standards can record data lineage [16]. Spec-driven AI development frameworks such as OpenSpec show a related move toward explicit proposals, requirements, designs, and task artifacts for coding agents [5], but they target software development workflows rather than executable decision-simulation worlds.

The specific gap is narrower: there is no clear workflow that converts a portfolio decision question into a validated WorldSpec and then connects evidence, assumptions, agent actions, state changes, metrics, and final report claims into one auditable chain. Current approaches can generate plausible scenarios, retrieve supporting documents, or log execution traces, but these pieces are rarely combined into a runnable and inspectable financial decision workflow that remains stable across many steps.

The key research problem is therefore not simply whether an LLM can imagine a scenario. It is whether a system can transform an ill-structured decision into an explicit, validated, executable, and inspectable artifact through staged orchestration. Compared with one-shot agent responses, this approach keeps evidence, assumptions, simulation state, agent actions, and final claims more clearly separated.

The proposal asks four research questions:

1. To what extent can a schema-based WorldSpec reduce invalid or incomplete portfolio scenarios compared with one-shot LLM scenario generation?
2. How effectively can evidence cards, assumption records, and provenance traces link final report claims to their supporting sources and simulation events?
3. Does a staged agentic workflow improve the clarity, inspectability, and decision usefulness of portfolio scenario reports compared with prompt-only and single-agent baselines?
4. What reliability costs are introduced by validation, repair, retrieval, and tracing, and which components contribute most to long-horizon workflow stability?

3 Proposed Method

The method is a staged workflow with explicit inputs, processing steps, and outputs. It is designed for portfolio scenario analysis, not general financial forecasting.

Problem structuring. The input is a natural-language portfolio question, such as whether to shift a moderate-risk portfolio toward equities under uncertain interest-rate and inflation conditions. The process extracts goals, constraints, candidate allocations, time horizon, risk tolerance, and evaluation metrics. The output is a DecisionSpec containing the decision objective, feasible interventions, and metrics such as expected return, drawdown, volatility, liquidity pressure, and capital exposure.

Evidence and assumption registration. The inputs are historical price data, market notes, and user-provided constraints. The process converts source-backed facts into EvidenceCards and unsupported modeling choices into editable Assumptions. The output is a small evidence and assumption registry that separates observed information from choices made for the simulation.

World-generation harness. The inputs are the DecisionSpec, EvidenceCards, and Assumptions. The process generates a constrained WorldSpec with asset groups, stakeholder agents, scenario parameters, interventions, update rules, metrics, and source links. Before execution, the harness checks schema validity, missing references, parameter ranges, unsupported claims, and executable constraints. This follows the simulation principle that verification and validation should be built into modeling [23]. The output is either a valid runnable WorldSpec or a repair request with explicit error messages.

Controlled orchestration. The inputs are validated or repairable artifacts from earlier stages. The process routes work through small modules: scenario drafting, validation, repair, compilation, execution, retrieval, tracing, and report generation. This design follows work on transparent chained

workflows [30] and graph-structured agentic workflow evaluation [20]. The output is an execution record in which each module has a defined responsibility and observable failure state.

Runtime, tracing, and reporting. The inputs are the compiled WorldSpec and selected scenarios. The runtime executes scenario steps with stakeholder agents representing an investor, analyst, and risk manager. At each step, agents observe current metrics, retrieve relevant evidence, take bounded actions, and update shared state. The output is a visual timeline, a machine-readable provenance trace, and a report whose claims must link to evidence, assumptions, and trace events, extending standard provenance toward agentic workflows [26].

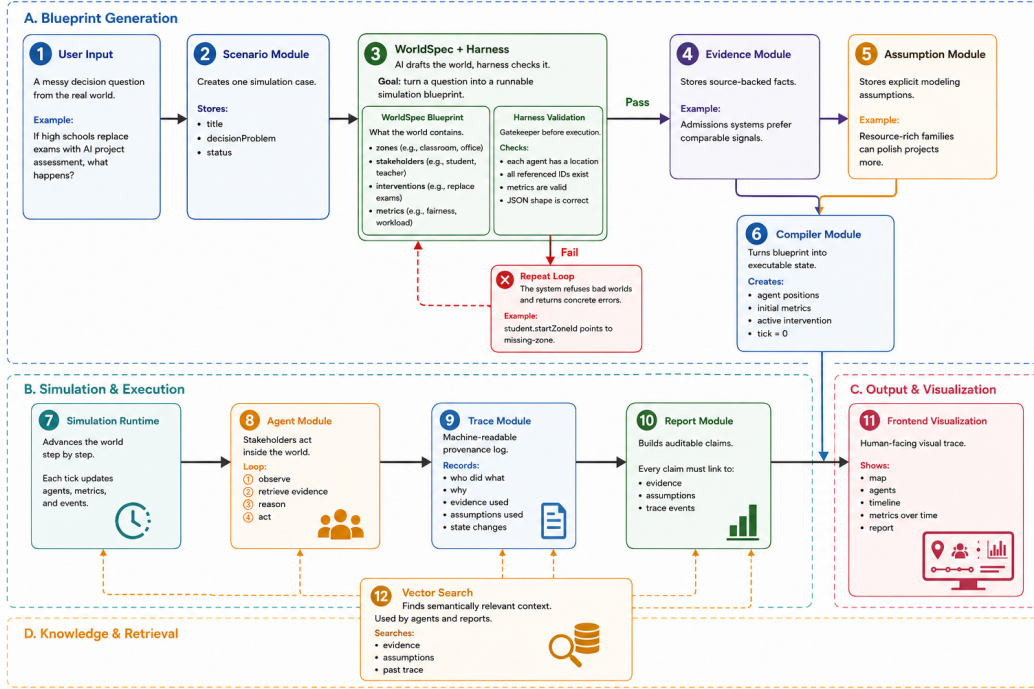


Figure 1: Harnessed agentic simulation architecture for financial scenario generation and traceable decision support. The workflow decomposes a financial decision question into scenario creation, world-specification drafting, harness validation, repair, evidence and assumption registration, compilation, stepwise simulation, agent action, provenance tracing, evidence-linked reporting, frontend visualization, and vector-search retrieval.

Figure 1 shows the concrete orchestration graph. Its main role is to force intermediate artifacts into the open: bad worlds are rejected before execution, missing evidence remains visible, assumptions are editable, and final claims require trace support. The core objects are explicit data structures rather than prompt text, making the system inspectable and testable.

Object	Required Fields	Purpose
WorldSpec	zones, agents, interventions, metrics, rules	Runnable simulation blueprint.
EvidenceCard Assumption	source, claim, scope, reliability statement, parameter, uncertainty, editable flag	Source-backed factual grounding. Explicit modeling choice.
TraceEvent	tick, actor, action, state change, links	Replayable provenance record.
ReportClaim	claim, evidence links, assumption links, trace links	Auditable conclusion.

Table 1: Core structured objects used by the harnessed workflow.

4 Main Example and Evaluation

The main case is a moderate-risk portfolio reallocation problem over a 12-month horizon. The prototype will use broad asset classes rather than individual stock picking: U.S. equities, aggregate bonds, gold, and cash or money-market exposure. The market scenarios will include a base case, an inflation-and-rate-pressure case, and an equity drawdown case. This level of detail keeps the case financially recognizable while avoiding unsupported claims about precise market prediction.

The system will compare three candidate allocations: a growth allocation with higher equity exposure, a balanced allocation, and a conservative allocation with more bonds, gold, and cash. Stakeholder agents represent an investor, an analyst, and a risk manager. Metrics include scenario return, maximum drawdown, volatility, liquidity pressure, and exposure concentration. The case is grounded in decision analysis, robust scenarios, and simulation-based reasoning [6, 9, 27]. An open-source demo already demonstrates the core pipeline and visualization layer [34].

Evaluation will compare four systems: a direct prompt-only LLM answer, a single tool-using LLM agent, the full harnessed workflow, and ablated workflow variants without validation, without retrieval, or without provenance tracing. The main quantitative measures are WorldSpec validity rate, consistency of intermediate state across steps, number of unsupported claims, percentage of report claims linked to evidence and trace events, repair-loop success rate, and runtime overhead. A small human evaluation will compare reports on clarity, inspectability, and usefulness for decision support, following recent work on realistic and fine-grained agent evaluation [13, 14, 21, 32, 33, 35].

5 Research Timeline

The project is scoped as an 18-month research plan. The goal is not to claim a general theory of all autonomous agents, but to deliver one well-specified orchestration framework, one runnable prototype, one detailed finance case study, and one focused evaluation.

Phase	Focus	Output
Months 1–3	Literature review; research questions; system scope.	Schema and requirements.
Months 4–6	Scenario, evidence, assumption, and WorldSpec modules.	Validated WorldSpec and repair-loop prototype.
Months 7–9	Compiler, runtime, agent loop, trace logging, and vector search.	Runnable finance scenario simulation.
Months 10–12	Frontend visualization and auditable report generation.	Visual timeline, metrics, traces, and linked report.
Months 13–15	Evaluation against prompt-only and single-agent baselines.	Traceability, validity, ablation, and error analysis.
Months 16–18	Framework refinement, thesis writing, and paper preparation.	Final thesis, reproducible demo, and paper draft.

Table 2: Planned research timeline and deliverables.

6 Expected Outcomes and Value

The expected outcome is not only a prototype, but reusable infrastructure for studying complex decisions with long-running agents. The project addresses a missing layer between flexible LLM agents and rigorous decision-support simulation: a structured workflow in which evidence, assumptions, intermediate states, and final claims remain inspectable.

The value is threefold. For AI-agent research, the project studies how validation, repair loops, provenance, retrieval, and modular orchestration can make long-running agents more reliable than one-shot answer generation. For decision-support research, it offers a way to compare scenarios under uncertainty without hiding the assumptions behind a recommendation. For interdisciplinary research, it provides a reusable workflow for domains where real-world trial and error is costly, slow, or ethically difficult.

Without such infrastructure, agentic decision support may remain fluent but hard to trust: users may receive plausible recommendations without knowing which evidence was used, which assumptions shaped the result, whether the simulated world was valid, or how errors propagated through the workflow.

The concrete outcomes are:

1. A compact orchestration schema that makes goals, evidence, assumptions, interventions, world specifications, intermediate states, and outcomes explicit.
2. A small but complete harnessed agentic simulation pipeline for scenario comparison under uncertainty.
3. A validation and repair framework for preventing AI-generated world specifications from entering execution when they are incomplete or invalid.
4. A dual trace layer that records human-facing visual traces and machine-readable provenance traces for auditability.
5. A focused case study in finance showing how traceable decision simulation can support practical discussion without relying on a single one-shot agent answer.
6. An open-source demo implementation that makes the proposed workflow reproducible and inspectable.

7 Related Work

LLM-agent research has advanced reasoning, acting, reflection, multi-agent collaboration, and agent evaluation [11, 13, 25, 29, 31]. Recent surveys further organize this area around autonomous-agent architectures, agentic LLM capabilities, and multi-agent workflow infrastructure [12, 19, 28]. Generative Agents and AI Town show how LLM-driven agents can inhabit persistent worlds [1, 18]. This proposal builds on these systems but shifts the goal from open-ended behavior to controlled, auditable scenario comparison.

The project also draws on agent-based modeling, robust decision-making, retrieval, and provenance. ABM explains how local interactions generate system-level effects [2–4, 15, 24], while robust scenario methods address decision-making under deep uncertainty [9, 27]. Retrieval-augmented generation grounds outputs in documents [10]; W3C PROV and PROV-AGENT support traceable links among activities, agents, prompts, decisions, and downstream effects [16, 26]. The gap is an integrated framework that connects these ideas into a validated agentic workflow for finance with evidence-linked reports.

8 Challenges

Three challenge-strategy pairs shape the project. First, generated simulations can look plausible without being decision-useful. The strategy is to avoid prediction claims and instead evaluate traceability, assumption visibility, and sensitivity under alternative scenarios [9, 22, 27]. Second, agentic workflows add orchestration cost. The strategy is to measure runtime overhead and use ablations to test whether validation, repair, retrieval, and tracing justify their cost. Third, the

framework may not generalize beyond the first case. The strategy is to begin with one concrete finance case, define the reusable schema clearly, and treat any later cross-domain use as an extension rather than an initial claim.

References

- [1] a16z Infra. AI Town: A virtual town where ai characters live, chat, and socialize. <https://github.com/a16z-infra/ai-town>, 2024. Accessed 2026-05-06.
- [2] Robert Axelrod. *The Complexity of Cooperation: Agent-Based Models of Competition and Collaboration*. Princeton University Press, 1997.
- [3] Eric Bonabeau. Agent-based modeling: Methods and techniques for simulating human systems. *Proceedings of the National Academy of Sciences*, 99(suppl_3):7280–7287, 2002.
- [4] Joshua M. Epstein and Robert Axtell. *Growing Artificial Societies: Social Science from the Bottom Up*. Brookings Institution Press, 1996.
- [5] Fission AI. OpenSpec: Spec-driven development for ai coding assistants. <https://github.com/Fission-AI/OpenSpec>, 2026. Accessed 2026-05-13.
- [6] Ronald A. Howard. Decision analysis: Applied decision theory. In *Proceedings of the Fourth International Conference on Operational Research*. Wiley-Interscience, 1966.
- [7] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):248, 2023. doi: 10.1145/3571730.
- [8] Peter G. W. Keen and Michael S. Scott Morton. *Decision Support Systems: An Organizational Perspective*. Addison-Wesley, Reading, MA, 1978.
- [9] Robert J. Lempert, David G. Groves, Steven W. Popper, and Steven C. Bankes. A general, analytic method for generating robust strategies and narrative scenarios. *Management Science*, 52(4):514–528, 2006. doi: 10.1287/mnsc.1050.0472.
- [10] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- [11] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for “mind” exploration of large scale language model society. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [12] Xinyi Li, Sai Wang, Siqi Zeng, Yu Wu, and Yi Yang. A survey on llm-based multi-agent systems: Workflow, infrastructure, and challenges. *Vicinagearth*, 1:9, 2024. doi: 10.1007/s44336-024-00009-2.
- [13] Xiao Liu et al. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, 2024.
- [14] Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan, Lingpeng Kong, and Junxian He. AgentBoard: An analytical evaluation board of multi-turn llm agents. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.
- [15] Charles M. Macal and Michael J. North. Tutorial on agent-based modelling and simulation. *Journal of Simulation*, 4(3):151–162, 2010. doi: 10.1057/jos.2010.3.
- [16] Luc Moreau and Paolo Missier. PROV-DM: The PROV data model. W3C recommendation, World Wide Web Consortium, 2013.

- [17] Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- [18] Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023. doi: 10.1145/3586183.3606763.
- [19] Aske Plaat, Max van Duijn, Niki van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. Agentic large language models, a survey. *Journal of Artificial Intelligence Research*, 84, 2025. doi: 10.1613/jair.1.18675.
- [20] Shuofei Qiao, Runnan Fang, Zhisong Qiu, Xiaobin Wang, Ningyu Zhang, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. Benchmarking agentic workflow generation. In *International Conference on Learning Representations*, 2025.
- [21] Yujia Qin et al. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *International Conference on Learning Representations*, 2024.
- [22] Andrea Saltelli, Karen Chan, and E. Marian Scott, editors. *Sensitivity Analysis*. John Wiley & Sons, Chichester, UK, 2000.
- [23] Robert G. Sargent. Verification and validation of simulation models. *Journal of Simulation*, 7 (1):12–24, 2013. doi: 10.1057/jos.2012.20.
- [24] Thomas C. Schelling. Dynamic models of segregation. *Journal of Mathematical Sociology*, 1 (2):143–186, 1971.
- [25] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [26] Renan Souza, Amal Gueroudji, Stephen DeWitt, Daniel Rosendo, Tirthankar Ghosal, Robert Ross, Prasanna Balaprakash, and Rafael Ferreira da Silva. PROV-AGENT: Unified provenance for tracking AI agent interactions in agentic workflows. In *IEEE International Conference on e-Science*, 2025.
- [27] Warren E. Walker, Robert J. Lempert, and Jan H. Kwakkel. Deep uncertainty. In Saul I. Gass and Michael C. Fu, editors, *Encyclopedia of Operations Research and Management Science*, pages 395–402. Springer, 3 edition, 2013.
- [28] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 18:186345, 2024. doi: 10.1007/s11704-024-40231-1.
- [29] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Shaokun Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Chi Zhang, Shaokang Liu, et al. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.
- [30] Tongshuang Wu, Michael Terry, and Carrie J. Cai. AI chains: Transparent and controllable human-AI interaction by chaining large language model prompts. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2022. doi: 10.1145/3491102.3517582.
- [31] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.
- [32] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik R. Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. In *International Conference on Learning Representations*, 2025.

- [33] Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel, Yilun Zhao, Roy Bar-Haim, Arman Cohan, and Michal Shmueli-Scheuer. Survey on evaluation of llm-based agents. *arXiv preprint arXiv:2503.16416*, 2025.
- [34] Cenxi Zhang. Traceable agentic simulation demo. <https://github.com/zcxixixi/traceable-agentic-simulation-demo>, 2026. Open-source prototype; accessed 2026-05-07.
- [35] Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, 2024.